**Example: The Ammonium Molecule**

**Question:**

Consider the two states $|1\rangle$ and $|2\rangle$ of the ammonium molecule outlined in the figure. Suppose they are orthonormal, $\langle i|j|i|j\rangle = \delta_{ij}$, and that only these two states are accessible to the system, so that we can describe it using the basis formed by $\{|1\rangle, |2\rangle\}$. On this basis the Hamiltonian of the system is given by

$$\hat{H} = \begin{pmatrix} E_0 & -E_1 \\ -E_1 & E_0 \end{pmatrix}.$$

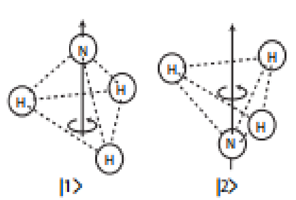
(a) If the system is initially in state $|1\rangle$, will it remain in that state at a later time? How about if the initial state is $|2\rangle$?

(b) Obtain the eigenvalues $E_I$ and $E_{II}$ and the respective eigenvectors $|I\rangle$ and $|II\rangle$ of $\hat{H}$, expressing them in terms of $|1\rangle$ and $|2\rangle$.

(c) What is the probability of measuring the energy $E_I$ in the following state,

$$|\psi\rangle = \frac{1}{\sqrt{5}}|1\rangle - \frac{2}{\sqrt{5}}|2\rangle\,?$$

(d) Based on the above result, predict at least one possible electromagnetic radiation emission frequency for an ammonium sample. What is this frequency?




The two states of the ammonium molecule.

**Solution:**

**(a) Time evolution of $|1\rangle$ and $|2\rangle$**

A state remains unchanged in time (up to a global phase) if and only if it is an eigenstate of the Hamiltonian.

Let us check whether $|1\rangle$ is an eigenstate:

$$\hat{H}|1\rangle = E_0|1\rangle - E_1|2\rangle.$$

Since this is not proportional to $|1\rangle$ alone (because of the $|2\rangle$ term), $|1\rangle$ is *not* an eigenstate. Therefore, if the system starts in $|1\rangle$, it will evolve into a superposition of $|1\rangle$ and $|2\rangle$.

Similarly,

$$\hat{H}|2\rangle = -E_1|1\rangle + E_0|2\rangle,$$

which is not proportional to $|2\rangle$. Hence $|2\rangle$ is also not an eigenstate, and the system will not remain in $|2\rangle$ either.

**Conclusion:** Neither $|1\rangle$ nor $|2\rangle$ are stationary states. The system oscillates between them.

**(b) Eigenvalues and Eigenvectors**

We solve the characteristic equation:

$$\det\left(\hat{H} - EI\right) = 0.$$

$$\det\begin{pmatrix} E_0 - E & -E_1 \\ -E_1 & E_0 - E \end{pmatrix} = (E_0 - E)^2 - E_1^2 = 0.$$

Thus,

$$(E_0 - E)^2 = E_1^2$$

$$E_0 - E = \pm E_1.$$

Hence the eigenvalues are:

$$E_I = E_0 - E_1, \tag{1}$$
$$E_{II} = E_0 + E_1. \tag{2}$$

**Eigenvectors**

For $E_I = E_0 - E_1$:

$$(\hat{H} - E_I I) = \begin{pmatrix} E_1 & -E_1 \\ -E_1 & E_1 \end{pmatrix}.$$

This gives the condition:

$$c_1 = c_2.$$

Normalized eigenvector:

$$|I\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle).$$

For $E_{II} = E_0 + E_1$:

$$(\hat{H} - E_{II} I) = \begin{pmatrix} -E_1 & -E_1 \\ -E_1 & -E_1 \end{pmatrix}.$$

This gives:

$$c_1 = -c_2.$$

Normalized eigenvector:

$$|II\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle).$$

**(c) Probability of measuring $E_I$**

We compute:

$$P(E_I) = |\langle I|\psi|I|\psi\rangle|^2.$$

$$\langle I|\psi|I|\psi\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{5}} - \frac{2}{\sqrt{5}}\right) = \frac{1}{\sqrt{2}}\left(\frac{-1}{\sqrt{5}}\right) = -\frac{1}{\sqrt{10}}.$$

Therefore,

$$P(E_I) = \left|-\frac{1}{\sqrt{10}}\right|^2 = \frac{1}{10}.$$

**(d) Emission frequency**

The possible emitted photon frequency corresponds to the energy difference:

$$\Delta E = E_{II} - E_I = (E_0 + E_1) - (E_0 - E_1) = 2E_1.$$

Using the Planck relation:

$$\Delta E = \hbar\omega,$$

we obtain:

$$\omega = \frac{2E_1}{\hbar}.$$

In terms of ordinary frequency,

$$\nu = \frac{\omega}{2\pi} = \frac{2E_1}{h}.$$

**Final Result:**

- The system oscillates between $|1\rangle$ and $|2\rangle$.
- Energy eigenstates are symmetric and antisymmetric superpositions.
- $P(E_I) = \frac{1}{10}$.
- Emission frequency: $\omega = \dfrac{2E_1}{\hbar}$.